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Encephalization and longevity evolved in a correlated fashion in Euarchontoglires but not in other mammals

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Across mammals, encephalization and longevity show a strong correlation. It is not clear, however, whether these traits evolved in a correlated fashion within mammalian orders, or when they do, whether one trait drives changes in the other. Here, we compared independent and correlated evolutionary models to identify instances of correlated evolution within six mammalian orders. In cases of correlated evolution, we subsequently examined transition patterns between small/large relative brain size and short/long lifespan. In four mammalian orders, these traits evolved independently. This may reflect constraints related to energy allocation, predation avoidance tactics, and reproductive strategies. Within both primates and rodents, and their parent clade Euarchontoglires, we found evidence for correlated evolution. In primates, transition patterns suggest relatively larger brains likely facilitated the evolution of long lifespans. Because larger brains prolong development and reduce fertility rates, they may be compensated for with longer lifespans. Furthermore, encephalization may enable cognitively-complex strategies that reduce extrinsic mortality. Rodents show an inverse pattern of correlated evolution, whereby long lifespans appear to have facilitated the evolution of relatively larger brains. This may be because longer lived organisms have more to gain from investment in encephalization. Together, our results provide evidence for the correlated evolution of encephalization and longevity, but only in some mammalian orders.

KEY WORDS: Cognitive buffer, delayed benefits, encephalization, expensive brain, longevity.

Across mammals, relatively large brains and incredibly long lifespans have evolved multiple times (Sacher 1959; Jerison 1973). This is surprising because brains that are larger than expected for a given body size are metabolically expensive (Aiello and Wheeler 1995; Isler and van Schaik, 2006, 2009a,b) and require extended developmental periods, which increase offspring mortality risk (Deaner et al. 2003; Barrickman et al. 2008) and delay age of first reproduction (Deaner et al. 2003; Barrickman et al. 2008). Extended longevity may also produce fitness costs, either due to antagonistic pleiotropy (i.e., mutations that allow a longer lifespan may have a detrimental effect on some aspect of early fitness; Williams 1957) or trade-offs between reproductive and maintenance functions (disposable-soma theory; Kirkwood 1979). Consequently, both encephalization and longevity should be favored by natural selection only when they confer fitness benefits that outweigh relevant growth and maintenance costs.

Interestingly, numerous studies have found a positive correlation between encephalization and longevity (across all mammals: Friedenthal 1910; Sacher 1959; Mallouk 1975; Isler and van Schaik 2009a,b; Gonzales-Lagos et al. 2010; primates: Austad and Fischer 1992; Hakeem et al. 1996; Deaner et al. 2003; Kaplan et al. 2003; Barrickman et al. 2008; Street et al. 2017; ungulates:

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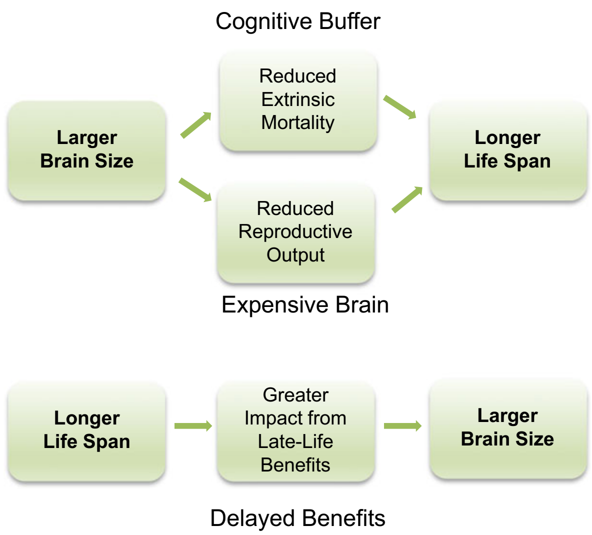

Figure 1. **Alternative hypotheses of causality in the correlated evolution of encephalization and longevity. The cognitive buffer hypothesis and expensive brain framework (top) suggest that greater encephalization reduces extrinsic mortality or reproductive output, leading to the evolution of longer lifespans. The delayed benefits hypothesis (bottom) suggests longer lifespans increase the benefits of information acquisition and behavioral flexibility, facilitating the evolution of greater encephalization.**

Cutler 1979; carnivores: Cutler 1979; Gittleman 1986). Consequently, researchers have put forth several hypotheses to explain why these traits appear to evolve in a correlated manner (Deaner et al. 2003). Two prominent hypotheses that offer an adaptive explanation for this relationship are the "delayed benefits" and "cognitive buffer" hypotheses. The former posits that long-lived species are more likely to benefit from relatively large brains because the associated cognitive abilities (e.g., memory, learning) provide benefits that accumulate throughout the lifetime, such that maximum potential payoff is reached later in life (i.e., the full realization of such benefits is "delayed"). More specifically, because longer lived organisms are more likely to encounter environmental changes, they gain greater benefits from information acquisition and behavioral flexibility (Deaner et al. 2003). Furthermore, long-lived animals have more time to learn and develop behaviors that can aid in their survival (Lefebvre et al. 2006). Longer lifespans should, therefore, increase the likelihood of evolutionary increases in relative brain size (Fig. 1; Kaplan et al. 2000; Deaner et al. 2003).

In contrast, the cognitive buffer hypothesis argues that encephalization provides more behavioral flexibility to respond to ecological challenges (e.g., predation, food shortages), thus reducing extrinsic mortality and allowing for the evolution of an extended lifespan. This idea has garnered support from studies showing that species with relatively large brains are more innovative (Reader and Laland 2002; Sol et al. 2005; Navarrete et al. 2016), more successful in novel environments (Sol et al. 2008), and exhibit lower net food intake variation in seasonal environments (van Woerden et al. 2012). According to this hypothesis, longer lifespans are, therefore, an evolutionary consequence of encephalization, rather than a cause (Fig. 1; Allman et al. 1993). An additional explanation for why encephalization may promote longevity is provided by the "expensive brain" framework. Because relatively larger brains are more energetically expensive, they tend to prolong development and limit reproductive rates. Consequently, longer lifespans may evolve to compensate for reduced reproductive output (Isler and van Schaik 2009a). Unfortunately, existing studies have not elucidated the applicability of these ideas since they have used correlation/regression analyses, which cannot imply directionality.

There are, however, a number of studies that do not find a correlation between encephalization and longevity (e.g., across all mammals: Mace 1979; Read and Harvey 1989; Austad and Fischer 1992; primates: Barton 1999; Ross et al. 1999; Judge and Carey 2000). Given that many mammal-wide studies use datasets that are dominated by primate species, it is possible that correlated evolution of longevity and encephalization is a pattern unique to primates, representing an "aberrant trend" among mammals (Austad and Fischer 1992). These biased datasets could mask patterns of independent evolution in relatively underrepresented orders, which might more accurately describe the typical mammalian trend (Austad and Fischer 1992). Accordingly, it may be inappropriate to extrapolate evolutionary patterns relevant to mammals in general from datasets that include a relatively high number of primate species. There are several scenarios in which one might expect encephalization and longevity to evolve independently. For example, relatively larger brains are associated with greater metabolic costs that may reduce an animal's ability to flee from predators, thereby increasing extrinsic mortality (Robin 1973; Hofman 1983). Similarly, the cognitive abilities afforded by encephalization may be translated into greater reproductive output instead of adult survival (Isler and van Schaik 2009a). Furthermore, species that evolve long lifespans may be subject to strong evolutionary constraints on encephalization, such as high energy allocation to locomotion (Pontzer and Wrangham 2004) or poor diet quality (Clutton-Brock and Harvey 1980; Martin 1983; DeCasien et al. 2017).

In the present study, we aimed to investigate the generality of the correlated evolutionary relationship between encephalization and longevity within multiple mammalian orders, including carnivorans, artiodactyls, cetaceans, chiropterans, primates, and rodents (**Aim 1**). In addition, for orders that demonstrated correlated evolution, we then examined evolutionary transition patterns between small/large relative brain size and short/long lifespans to illuminate their causal relationship and evaluate the relevance of different explanatory hypotheses (**Aim 2;** Fig. 1).

Methods and Materials

DATA COLLECTION

Data on maximum lifespan, species average brain weight, and species average body weight were collected for more than 600 mammalian species, including 151 carnivorans (Carnivora), 77 even-toed ungulates (Artiodactyla), 37 cetaceans (Cetacea), and 54 chiropterans (Chiroptera) among Laurasiatheria, and 168 rodents (Rodentia), 10 lagomorphs (Lagomorpha), three tree shrews (Scandentia), and 144 primates (Primates) among Euarchontoglires (see Supplementary Dataset 1). Maximum lifespan data were collected from multiple published sources (De Magalhaes and Costa 2009; Jones et al. 2009; Isler and van Schaik 2012; Supplementary Dataset 1), including both wild and captive records. Brain and body weights were collected from multiple published sources and represent averages of those reported values (Hutcheon et al. 2002; Jones and MacLarnon 2004; De Magalhaes and Costa 2009; Jones et al. 2009; Shultz and Dunbar 2010; Boddy et al. 2012; Isler and van Schaik 2012; Fox et al. 2017; primates: Isler et al. 2008; DeCasien et al. 2017; Supplementary Dataset 1). Given that information on sex is not available for the majority of our sources, these data are not sex specific.

Inferred phylogenies are subject to both stochastic and systematic errors, resulting in phylogenetic uncertainty that may influence results (Huelsenbeck et al. 2000). To consider the possible effects of such uncertainty, we incorporated multiple phylogenies for each mammalian group throughout the analyses described below, in line with previous work (e.g., Symonds and Elgar 2002; DeCasien et al. 2017). For carnivorans, we ran analyses using the 10KTrees consensus tree ($N = 144$ species; Version 1; Arnold et al. 2010) and a recent phylogenetic tree created using supertree analysis ($N = 149$ species; Nyakatura and Bininda-Emonds 2012). For cetaceans, we used the 10KTrees consensus tree ($N = 37$ species; Version 1; Arnold et al. 2010) and a composite of published molecular trees ($N = 27$ species; McGowen et al. 2009; McGowen 2011) that was aggregated by Montgomery and colleagues (Montgomery et al. 2013). For chiropterans, we ran analyses using a subset of a larger mammal-wide phylogeny ($N = 53$ species; Bininda-Emonds et al. 2008) and the recent tree produced by Agnarsson and colleagues ($N = 41$ species; 2011). For artiodactyls, we used the 10KTrees consensus tree ($N = 72$ species; Version 1; Arnold et al. 2010) and a subset of a mammal-wide phylogeny ($N = 77$ species; Bininda-Emonds et al. 2008). For primates, we ran analyses using the 10KTrees consensus tree ($N = 140$ species; Version 3; Arnold et al. 2010) and the tree produced by Perelman and colleagues ($N = 104$ species; Perelman et al. 2011). For rodents, we used the tree produced by Fabre and colleagues ($N = 157$ species; Fabre et al. 2012) and a subset of a mammal-wide phylogeny ($N = 168$ species; Bininda-Emonds et al. 2008). Post hoc analyses of the clade Euarchontoglires (primates, rodents, lagomorphs, and tree shrews) used a subset of a mammal-wide phylogeny ($N = 129$ primates, $N = 168$ rodents, $N = 3$ tree shrews, $N = 10$ lagomorphs; Bininda-Emonds et al. 2008). Lagomorphs and tree shrews were only included in analyses of Euarchontoglires.

STATISTICAL ANALYSES

All statistical analyses were carried out in R v 3.4.0 (R Core Team 2017) and BayesTraits V2 (Pagel and Meade 2013). All variables were log transformed prior to running phylogenetic generalized least squares (PGLS) regression models to decrease skewedness.

AIM 1: TESTING FOR CORRELATED EVOLUTION BETWEEN RELATIVE BRAIN SIZE AND LIFESPAN

The statistical analysis required for hypothesis testing (DISCRETE; Pagel 1994) requires that traits are categorical, so within each taxonomic group, we categorized each species as small or large brained and short or long lived, relative to the other species within the same taxonomic group (Perez-Barberia et al. 2007). Given that there is a strong positive relationship between brain and body size across mammals and within orders (Table S1; Sacher 1959; Jerison 1973), we calculated each species' relative brain size as its residual from a PGLS regression of brain size (log) on body size (log) within each order/clade. Individual species were categorized as "large-brained" (= 1) if their brain size residual value was positive or "small-brained" (= 0) if their brain size residual value was negative. Because of the lack of an equally strong relationship between lifespan and body size within these orders (Table S1), and the fact that relative brain size is better predicted by absolute, rather than relative, lifespan (Gonzalez-Lagos et al. 2010), we characterized each species as "long-lived" (= 1) if lifespan was greater than and "short-lived" (= 0) if lifespan was less than their order's geometric mean value. To address the fact that both lifespan and relative brain size are continuous variables with distributions that vary by order, we followed prior studies (Fristoe et al. 2017) and implemented a classification scheme in which we categorized species as "long-lived" or "large-brained" according to different thresholds, that is $\geq 30^{th}$, $\geq 50^{th}$, and $\geq 75^{th}$ percentiles of the lifespan and residual brain size distributions. We assessed the relationships between categorical brain size and life span for all cutoff combinations, incorporating the phylogeny that provided the highest species sample size. Species were also categorized in this manner within the larger clade Euarchontoglires. We acknowledge that the skewed distribution toward small/large brain sizes in some groups is due to the effects of phylogenetic correction and the resulting multivariate normal error distribution (Freckleton et al. 2011).

We used the categorical measures of relative brain size and lifespan to classify species into 2×2 contingency tables for each

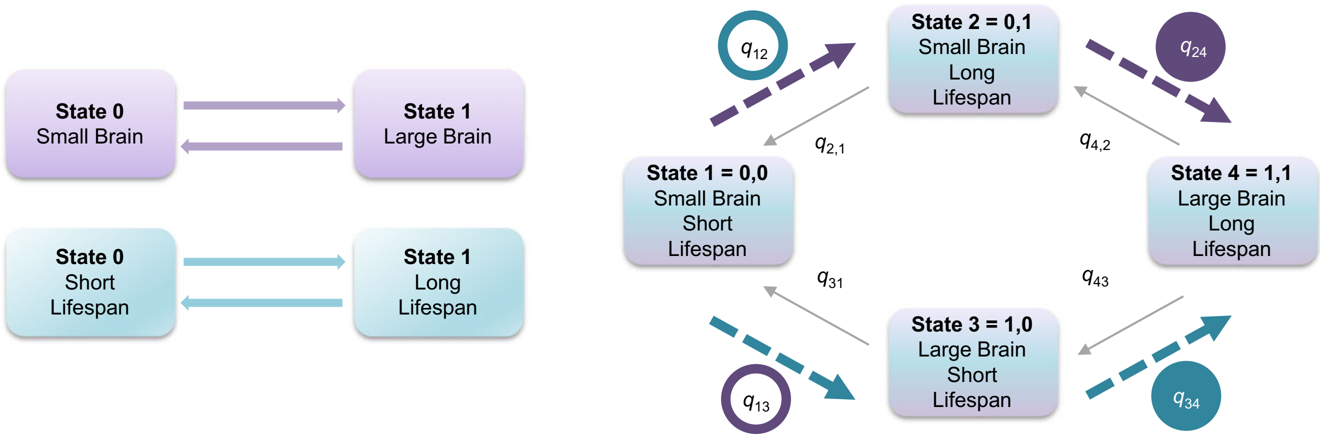

Figure 2. Pagel's (1994) discrete variables compare the likelihood of independent (left) versus correlated (right) evolution of two binary traits. The independent evolution model estimates four parameters, and the correlated evolution model estimates eight transition rate (q_{ij}) parameters (since the two variables change in a correlated fashion across the tree). Transition rate patterns that provide evidence for correlated evolution in accordance with either the delayed benefits hypothesis (purple bold arrows) or the cognitive buffer and expensive brain hypotheses (teal bold arrows) are also depicted. The transition rates being compared in MCMC analyses are circled, such that a filled circle highlights the transition rate that is expected to be relatively larger than the transition rate in the hollow circle, in accordance with a given hypothesis (colors correspond to the hypotheses as listed above).

taxonomic group and used Fisher's exact tests to evaluate the distribution of small/large brained and short/long lived species within each group (Perez-Barberia et al. 2007).

We tested if there was evidence for correlated evolution of relative brain size and lifespan within each order by comparing the likelihood of independent and correlated models of evolution using Pagel's discrete variables method (Fig. 2; Pagel 1994). This method uses a continuous-time Markov model to characterize evolutionary changes along each branch of a phylogenetic tree. In the first model (independent evolution), the two binary values of relative brain size and lifespan are allowed to change independently across the branches of the tree. Each of the two variables can go through two evolutionary state transitions ($0 \rightarrow 1$; $0 \leftarrow 1$), so this model estimates four parameters (i.e., the transition rates of relative brain size from small to large and large to small, and also the two rates for lifespan). In the second model (correlated evolution), the two variables change in a correlated fashion across the tree. Four possible states of two combined binary values are possible (State 1 = 0,0; State 2 = 0,1; State 3 = 1,0; State 4 = 1,1; Fig. 2), and each transition between these states is defined by a transition rate parameter (q_{ij}) that represents the rate of change from state i to state j (e.g., q_{12} = the rate of change from state 1 $\rightarrow$ state 2 or 0,0 $\rightarrow$ 0,1). Allowing either relative brain size or lifespan to change state within any branch of the tree produces eight possible linked-character transitions, so this model estimates eight transition rate (q_{ij}) parameters. We compared the fit of the correlated evolution model, *Likelihood*(C), to the independent evolution model, $L(I)$, using a likelihood ratio statistic $LR = -2 \ln [L(I)/L(C)]$ and a chi-squared test with four degrees of freedom (equal to the difference in the number of parameters between the two models). If $L(C)$ was not significantly larger than $L(I)$ ($P > 0.05$), we concluded that there was no evidence for correlated evolution between relative brain size and lifespan; however, if $L(C)$ was significantly larger than $L(I)$ ($P < 0.05$), we accepted that there was correlated evolution between these traits. We also examined differences in Akaike Information Criteria (dAIC) between correlated and independent models of evolution. Absolute dAIC values greater than or equal to 7 are considered evidence that the model with the lower AIC value exhibits significantly better model fit (Burnham et al. 2011).

Correlated evolution analyses were executed in R using the fitPagel function in the phytools package (Revell 2012). All three available optimization methods were used and checked for consistency (fitMk; ace; fitDiscrete). Prior to analyses, polytomies (if present) were resolved via transformation into a series of dichotomies with one or several branches of length zero (multi2di function, ape package: Paradis et al. 2012), and then a trivial branch length (1e-08) was added across the tree to avoid function errors. Likelihood analyses were also replicated in BayesTraits (Pagel and Meade 2013) without the aforementioned tree modifications.

AIM 2: ASSESSING PATTERNS OF CORRELATED EVOLUTIONARY TRANSITIONS

In cases of identified correlated evolution, we then went on to use the discrete variables method to test the temporal ordering and direction of evolutionary change of relative brain size and lifespan. This was executed using maximum likelihood analysis in BayesTraits V2 (Pagel and Meade 2013). The method identifies which are the more likely linked-character transitions by testing whether the likelihood of particular transitions is significantly different from zero. In doing so, the method allows one to make inferences about the potential direction and primary driver of the overall correlation. We tested models of evolution in which certain types of transitions are excluded a priori by separately constraining each transition rate parameter (q_{ij}) to zero (restrict function). The fit of each constrained model, in which one parameter is constrained to zero, is then compared to the full (unconstrained) model of eight parameters by comparing the log-likelihoods using a likelihood ratio statistic (as described above). If the fit of the unconstrained model is significantly better than that of a constrained model, the transition rate under manipulation is considered significantly greater than zero (Pérez-Barbería et al. 2007). The number of iterations per tree was set to 200 (mltries function) to produce stable likelihood values. Analyses were repeated with the root node set to reflect a relatively small brained, short-lived species (state = 0,0; fo function; as in Pérez-Barbería et al. 2007). Figure 2 depicts the patterns of significance expected if the delayed benefits hypothesis and the cognitive buffer or expensive brain hypotheses are relevant to the majority of evolutionary changes within a group.

For orders exhibiting correlated evolution, we also tested whether: (1) small to large brain transitions were more prevalent when starting from a long versus short lifespan (i.e., $q_{24} > q_{13}$; represents a test of the delayed benefits hypothesis; Fig. 2); and (2) short- to long-lifespan transitions were more prevalent when starting from a large versus small brain (i.e., $q_{34} > q_{12}$; represents a test of the cognitive buffer/expensive brain hypotheses; Fig. 2). Evolutionary transition rates were estimated using Markov Chain Monte Carlo (MCMC) methods in BayesTraits V2 (Pagel and Meade 2013). Following similar analyses (Fristoe et al. 2017), we used exponential priors whose mean is seeded from a uniform hyper-prior ranging between values relevant to each group. As recommended in the BayesTraits manual, parameter values were first estimated using maximum likelihood analysis to inform our choice of priors (Pagel and Meade 2013). Consequently, the ranges of our hyper-priors were 0–1.5 for primates, 0–1 for rodents, and 0–0.05 for Euarchontoglires. The hyper-prior is a distribution from which values are drawn to seed the values of exponential or gamma priors. This approach is recommended because it can reduce some of the "uncertainty and arbitrariness" of choosing priors in MCMC analysis (Pagel and Meade 2013). Exponential priors are recommended when smaller values are more likely than larger ones. For Euarchontoglires, MCMC analyses were run for 10,000,000 iterations with a burn-in of 100,000 and a thinning interval of 1000 iterations. For primates and rodents, MCMC analyses took longer to converge, so they were run for 100,000,000 iterations with a burn-in of 5,000,000, and a thinning interval of 10,000 iterations. At each iteration, we calculated the difference between relevant transition rates (i.e., q_{24}–q_{13}; q_{34}–q_{12}). To test if there was a significant, positive difference between the relevant transition rates, we calculated pMCMC values as the proportion of iterations in which the difference produced a negative value. We ran each MCMC analysis three times to ensure chain convergence and assess the consistency of our results. These checks were performed using the R package coda (Plummer et al. 2006). We visually inspected the traces of all of our posterior estimates, confirmed effective sample sizes were greater than 200, and used Gelman and Rubin's convergence diagnostic (Gelman and Rubin 1992) to assure PSRF values were below 1.1 for all parameter estimates.

Results

The orders included in this study exhibit extensive variation with regard to brain size, body size, relative brain size, and lifespan (Fig. S1). The relationship between lifespan and relative brain size also varies across groups, with rodents and primates exhibiting a strong positive relationship between these variables (Fig. 3). Overall, we find evidence for correlated evolution within primates and rodents only, and that transition patterns differ between these groups. Given that these orders are members of the parent clade Euarchontoglires, we ran post hoc tests on this larger clade to examine broader correlated evolutionary patterns. For each group, we present results in detail from the phylogeny that provided the larger sample size (see Supplementary Material for additional results). In addition, we present results from correlated evolution analyses using the "ace" function because this optimization method is most similar to that used in BayesTraits, and also provides model AIC values (see Methods and materials).

AIM 1: INSTANCES OF INDEPENDENT VERSUS CORRELATED EVOLUTION

Fisher's exact tests indicated a significant association between relative brain size and lifespan in primates, rodents, and in some analyses of carnivorans and artiodactyls (Tables 1, S2). Most carnivorans (>30%), artiodactyls (>40%), and primates (>40%) exhibit large brains and long lifespans. Rodents exhibit the opposite pattern, with most species (>40%) falling in the small brain and short lifespan category. The primate and rodent results are robust to species sampling and different cutoff values (Tables 1, S2–S3). Post hoc analyses of Euarchontoglires also suggest a significant association between brain size and lifespan in this clade (Table 1).

According to comparisons of the log-likelihoods of independent and correlated models of evolution, evidence for correlated evolution of encephalization and longevity exists only in primates, rodents, and their parent clade Euarchontoglires (Table 2). For all other orders, the log-likelihood of the correlated model was only

Figure 3. The relationship between lifespan and relative brain size varies across orders. Relative brain size represents residuals from a phylogenetic generalized least squares regression of brain size regressed on body size across all species. We used topologies and branch lengths from a mammal-wide phylogeny (Bininda-Emonds et al. 2008), and allowed the phylogenetic scaling factor (lambda) to take the value of its maximum likelihood. Shaded ovals represent normal confidence ellipses for each group.

Table 1. Contingency tables and Fisher's exact test results showing the distribution of species across relative brain size and lifespan categories.

Order	Phylogeny	*N*	Lifespan	Small brain	Large brain	Odd's ratio	*P*-value
Carnivora	Nyakatura and Bininda-Emonds 2012	149	Short	31	39	1.11	0.87
			Long	33	46		
Cetacea	Arnold et al. 2010	37	Short	1	15	0.14	0.10
			Long	7	14		
Chiroptera	Bininda-Emonds et al. 2008	53	Short	9	16	0.49	0.27
			Long	15	13		
Artiodactyla	Bininda-Emonds et al. 2008	77	Short	13	22	3.49	**0.03**
			Long	6	36		
Primates	Arnold et al. 2010	140	Short	34	27	12.67	**<0.01**
			Long	7	72		
Rodentia	Bininda-Emonds et al. 2008	168	Short	71	14	6.87	**<0.01**
			Long	35	48		
Euarchontoglires	Bininda-Emonds et al. 2008	310	Short	133	18	42.78	**<0.01**
			Long	23	136		

**P*-values represent Fischer's exact tests. Significant values are in bold.

slightly higher than the log-likelihood of the independent model, resulting in a small likelihood-ratio value, a nonsignificant chi-squared test value, and a low (<7) dAIC value. These results were consistent across optimization methods, phylogenies (i.e., they are robust to phylogenetic uncertainty), and different cutoff values for trait categories (Tables 2, S4–S5).

AIM 2: CORRELATED EVOLUTIONARY TRANSITION PATTERNS

Given that there was evidence for correlated evolution within primates and rodents, we examined the directions of evolutionary transitions in brain size and lifespan for each of these orders, and also their parent clade Euarchontoglires, by testing whether the

Table 2. Comparisons of independent and correlated models of evolution using Pagel's discrete method (1994).

Clade	Phylogeny	L(I)*	L(C)*	LR*	*P*-value†	dAIC‡
Carnivora	Nyakatura and Bininda-Emonds 2012	–155.69	–152.71	5.96	0.20	2.0
Cetacea	Arnold et al. 2010	–35.34	–30.97	8.75	0.07	–0.8
Chiroptera	Bininda-Emonds et al. 2008	–42.06	–40.75	2.63	0.62	5.4
Artiodactyla	Bininda-Emonds et al. 2008	–83.66	–80.42	6.47	0.17	1.5
Primates	Arnold et al. 2010	–126.15	–114.69	22.92	**<0.01**	**–14.9**
Rodentia	Bininda-Emonds et al. 2008	–185.78	–175.70	20.14	**<0.01**	**–12.1**
Euarchontoglires	Bininda-Emonds et al. 2008	–161.47	–145.65	31.65	**<0.01**	**–23.6**

*L(I) = likelihood of independent model; L(C) = likelihood of correlated model; LR = likelihood-ratio.

†*P*-values represent Chi-Squared tests with 4 degrees of freedom. Significant values are in bold.

‡dAIC values represent the difference between the estimated Akaike Information Criteria (AIC) for correlated and independent models. Negative values indicate lower AIC values for correlated models (i.e., correlated models are better fit). Absolute values greater than or equal to 7 are considered evidence for a significant difference in model fit and are in bold (Burnham et al. 2011).

Table 3. Likelihood comparisons of unconstrained correlated evolution models (UCM) versus constrained correlated evolution models (CCM) in which one of the transitions between states has been forced to be zero.

Order	L(UCM)*	Alternative models	L(CCM)*	LR*	*P*-value†
Primates	–114.69	$q_{12} = 0$	–114.69	0.00	0.996
		$q_{13} = 0$	–119.87	10.35	**0.001**
		$q_{21} = 0$	–119.35	9.32	**0.002**
		$q_{24} = 0$	–114.69	0.00	0.999
		$q_{31} = 0$	–116.78	4.18	**0.041**
		$q_{34} = 0$	–123.72	18.05	**<0.001**
		$q_{42} = 0$	–118.44	7.49	**0.006**
		$q_{43} = 0$	–126.57	23.74	**<0.001**
Rodentia	–175.82	$q_{12} = 0$	–183.93	16.21	**<0.001**
		$q_{13} = 0$	–175.74	0.16	0.693
		$q_{21} = 0$	–178.94	6.24	**0.012**
		$q_{24} = 0$	–178.62	5.60	**0.018**
		$q_{31} = 0$	–175.76	0.13	0.723
		$q_{34} = 0$	–177.28	2.91	0.088
		$q_{42} = 0$	–179.67	7.69	**0.006**
		$q_{43} = 0$	–181.60	11.55	**0.001**
Euarchontoglires	–145.65	$q_{12} = 0$	–165.72	40.15	**<0.001**
		$q_{13} = 0$	–153.89	16.49	**<0.001**
		$q_{21} = 0$	–149.03	6.76	**0.009**
		$q_{24} = 0$	–153.50	15.71	**<0.001**
		$q_{31} = 0$	–154.51	17.73	**<0.001**
		$q_{34} = 0$	–145.64	0.01	0.922
		$q_{42} = 0$	–147.03	2.76	0.097
		$q_{43} = 0$	–152.05	12.81	**<0.001**

Throughout, significant values indicate that the relevant alternative/constrained model is significantly worse than the unconstrained model, suggesting that the transition rate being forced to 0 is greater than 0. State 1 = small brain, short lifespan; State 2 = small brain, long lifespan; State 3 = large brain, short lifespan; State 4 = large brain, long lifespan. Models/transition rates supporting the cognitive buffer hypothesis are in teal, and those supporting the delayed benefits hypotheses are in purple.

*L(UCM) = likelihood of unconstrained correlated model with eight estimated parameters (these models are also in Table 2); L(CCM) = likelihood of constrained correlated model; LR = likelihood-ratio

†*P*-values represent Chi-Squared Tests with 1 degree of freedom. Significant values are in bold.

rate of either forward or backward changes through any of the intermediate states (i.e., State 2: small brain, long life; State 3: large brain, short life) was significantly different from zero. All groups exhibited significant backward and forward transitions from both intermediate states (i.e., to and from States 2 and 3; Fig. S2; Table 3). Primates and rodents both exhibited some degree of flexibility in terms of decoupling the association between relative brain size and lifespan, as they both exhibit significant transitions from relatively large brained, long lived species (State 4) to either relatively small brained, long-lived species (State 2) or relatively large brained, short-lived species (State 3; Fig. S2; Table 3). However, each order exhibited only one significant pathway by which relatively small brained, short-lived species (State 1) gave rise to large brained, long-lived species (State 4), and this evolutionary path differed in primates versus rodents.

In primates, analyses using our standard cutoffs suggest State 1 only leads to State 3, and State 4 only emerged from State 3 (Fig. S2; Tables 3, S6), indicating that relatively large brained, long-lived primate species evolved from species that already had relatively large brains. This result is consistent across phylogenies and regardless of whether or not the root node was set to State 1. Only two of nine analyses using alternative cutoff combinations suggested that State 4 exclusively emerged from State 2 in primates, rather than from state 3 (Table S7). In both cases, species classified as large-brained were those with the most extreme relative brain size, as its cutoff was set to 75%. This cut off resulted in 64 species (46% of all species in the analysis) with positive brain size residuals being classified as small-brained (compared to our standard cutoffs, this represents a ∼156% increase in the number of small-brained species, versus a 71% increase with a 50% cutoff; Tables S2–S3).

In rodents, most analyses using our standard cutoffs suggest State 1 only leads to State 2, and State 4 only emerged from State 2 (Fig. S2; Tables 3, S6), indicating that relatively large brained, long-lived rodent species evolved from species that already had long lifespans. These results vary slightly across phylogenies and root node setting methods, suggesting there is some uncertainty surrounding whether or not the alternative pathway is also significant (Table S6). Three of nine analyses using alternative cutoff combinations suggest that State 4 emerges exclusively from State 3 in rodents (Table S7). This transition pathway occurred exclusively when only extremely long lifespans were classified as long, with a cutoff set at 75%. This cutoff resulted in a 48% decrease in the number of species classified as long-lived (vs a 2% increase with a 50% cutoff; Tables S2–S3). Since the rodent lifespan distribution is skewed rightward, a 75% cutoff causes only species with very long lifespans to be classified as long-lived (i.e., mole rats and squirrels) and a large decrease (51%) in the overall number of transitions detected (Fristoe et al. 2017). All other cutoff combinations support our main findings or suggest both pathways are significant (Table S7). Across Euarchontoglires, State 1 leads to both States 2 and 3, but State 4 only emerges via State 2 (Fig. S2; Tables 3, S6).

For Primates, Rodentia, and Euarchontoglires, we also used MCMC analyses to test whether: (1) small to large brain transitions were more prevalent when starting from a long versus short lifespan (a test of the delayed benefits hypothesis); and/or (2) short to long lifespan transitions were more prevalent when starting from a large versus small brain (a test of the cognitive buffer/expensive tissue hypotheses). In primates, transitions from short to long lifespans occurred more often from a large brain state ($q_{34} > q_{12}$: pMCMC = 0.012; Fig. 4), while transitions from small to large brains did not occur more often from either short or long lifespan states ($q_{24} > q_{13}$: pMCMC = 0.187; Fig. 4). In rodents, analyses did not indicate a significant difference between any of the relevant transition rates ($q_{34} > q_{12}$: pMCMC = 0.068; $q_{24} > q_{13}$: pMCMC = 0.089; Fig. 4). In Euarchontoglires, analyses suggest transitions from small to large brains occurred more often from a long lifespan state ($q_{24} > q_{13}$: pMCMC = 0.028; Fig. 4), while transitions from short to long lifespans did not occur more often from either small or large brain states ($q_{34} > q_{12}$: pMCMC = 0.245; Fig. 4).

Discussion

For the majority of mammalian orders examined – including carnivorans, artiodactyls, cetaceans, and chiropterans (i.e., members of the major clade Laurasiatheria) – we find no evidence for correlated evolution of encephalization and longevity (Tables 2, S4–S5). This suggests that correlated evolution of these traits may not be the typical mammalian trend (Austad and Fischer 1992). Uniquely, members of Euarchontoglires do exhibit evidence of a correlated evolutionary relationship (Tables 2, S4–S5); however, transition patterns appear to differ in primates and rodents (Tables 3, S6–S7), suggesting different explanatory hypotheses may be relevant to each group. For primates, transition patterns suggest that: (1) relatively large brained, long-lived species tend to evolve from species that already have relatively large brains (Fig. S2; Tables 3, S6–S7); and correspondingly, (2) short to long lifespan transitions are more likely to occur in the presence of relatively large brains (Fig. 4). Within rodents, transition patterns suggest that relatively large brained, long-lived species tend to evolve from species that already exhibit long lifespans (Fig. S2; Tables 3, S6–S7). For Euarchontoglires overall, transition patterns are similar to the rodent results (Fig. 4; Tables 3, S6), likely reflecting the fact that the majority of species within this group are rodents (54%).

Given that encephalization and longevity appear to have evolved independently in multiple mammalian orders, these traits may often be uncoupled by constraints and tradeoffs,

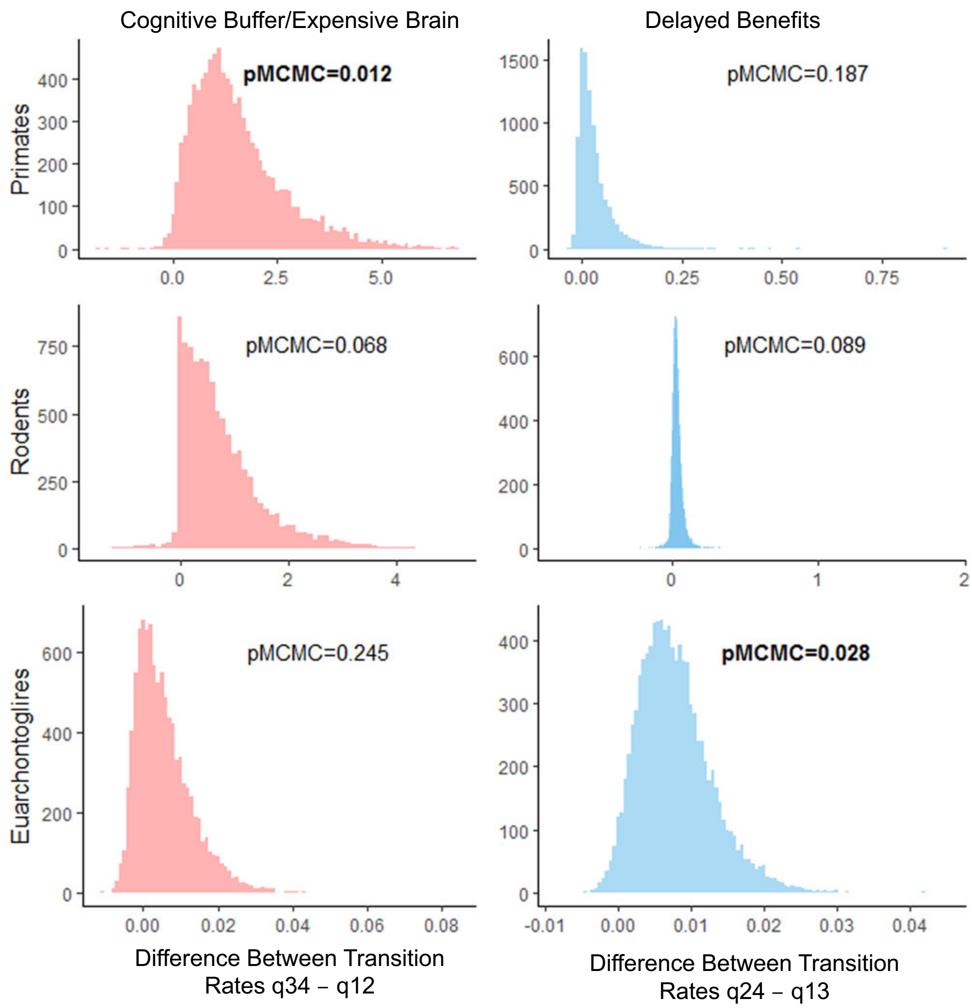

Figure 4. Posterior distributions of the difference in transition rate from short to long lifespans between small and large brain states (left panel; q_{34}–q_{12}) and from small to large brains between short and long lifespan states (right panel; q_{24}–q_{13}). Results from the first MCMC chain for each analysis are depicted. Significant pMCMC values (pMCMC < 0.05) are in bold and suggest a significant difference between the transition rates under examination.

including those associated with energy allocation, predation avoidance tactics, and reproductive strategies. For example, specializations to low quality diets may place energetic constraints on relative brain size, as in primates where diet quality is positively correlated with encephalization (DeCasien et al. 2017; Fig. 5). This may partially explain our results suggesting independent evolution within artiodactyls since most species have diets that are mainly composed of grasses or browse (Pérez-Barbería and Gordon 2005), both of which are less energy dense and require more energy to digest than fruit or meat. Additional factors affecting metabolism, such as survival strategies like hibernation, may also prevent the correlated evolution of encephalization and longevity. Hibernating mammals have longer lifespans due to reduced predation and higher survival rates during periods of hibernation (Turbill et al. 2011). Hibernation also allows calorie restriction, the only experimental method proven to increase lifespan in mammals (Lin et al. 2002). Furthermore, although bats have high metabolic rates compared to other mammals, their metabolic rate during hibernation is only about 5% of its resting metabolic rate (Wilkinson and South 2002). This may reduce the accumulation of intracellular oxidative damage, also promoting longer lifespans (Austad 1997; Wilkinson and South 2002; Munshi-South and Wilkinson 2010). However, given that metabolic rate is positively correlated with relative brain size in mammals (Isler and van Schaik 2006), indicating a link between energy turnover and the high energetic needs of brain tissue (Armstrong 1983), hibernation may constrain encephalization in long-lived species (Fig. 5). In addition, certain locomotor behaviors, such as volancy, might create additional energy allocation constraints that likewise uncouple encephalization and longevity. Volant species may flee from predators more easily, thereby reducing predation risk and increasing average lifespan (Pomeroy 1990; Healy et al. 2014). Accordingly,

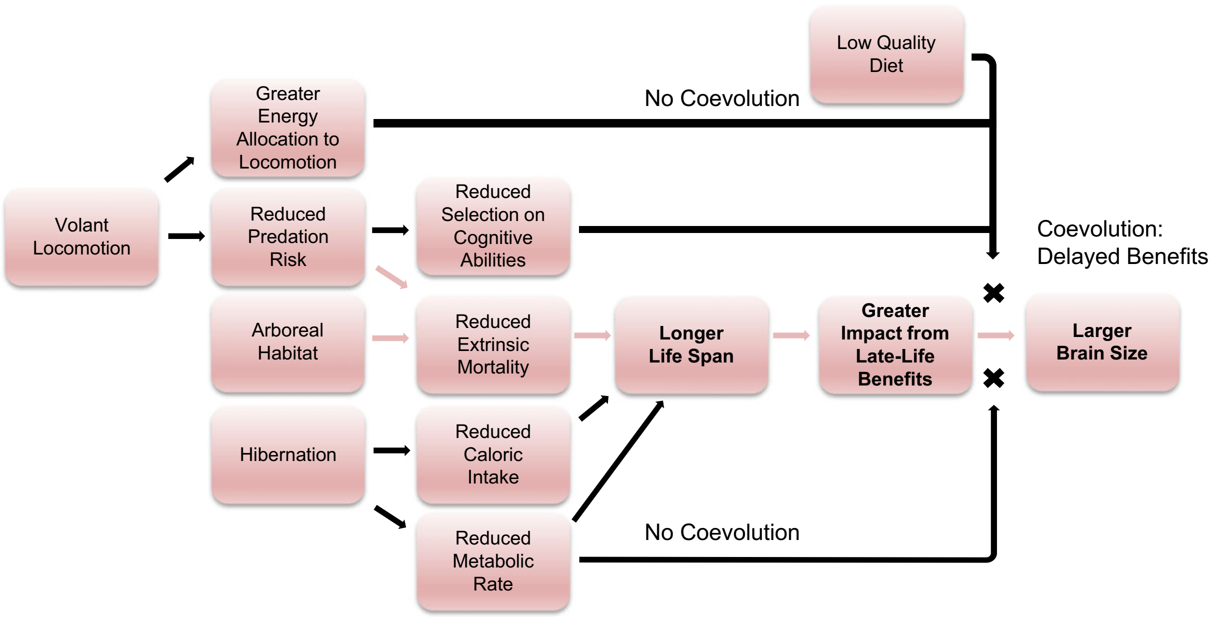

Figure 5. **Various factors can inhibit (in black) the emergence of relatively large brains in long-lived species, including energy allocation to locomotion, low quality diets, and metabolic restraints.**

bats live about 3.5 times longer than similarly sized nonflying placental mammals (Wilkinson and South 2002). Volancy is, however, an energetically expensive form of locomotion, and this may constrain relative brain size since encephalization is likely promoted by reductions in energy allocation to locomotion (Pontzer and Wrangham 2004; Isler and van Schaik 2006; Navarrete et al. 2011; Fig. 5). Moreover, the ability to escape predators may also remove selective pressure on cognitive abilities and, therefore, encephalization (Fig. 5).

Other forms of locomotion may also prevent the correlated evolution of encephalization and longevity due to predation-related constraints. More specifically, relatively larger brains may pose important biological handicaps on diving mammals since they require a larger constant supply of oxygen, and this supply is negatively influenced by longer dive times ("dive constraint hypothesis": Robin 1973; Hofman 1983). Consistent with this idea, encephalization is negatively correlated with maximum dive duration in diving mammals (Robin 1973), including cetaceans (Marino et al. 2006) and carnivorans pinnipeds (Fitzpatrick et al. 2012). Increased dive time is likely advantageous in avoiding predators (Robin 1973); therefore, relatively larger brain size in diving animals could lead to shorter maximum dive durations, increased risk of predation, greater extrinsic mortality, and, consequently, a constraint on longevity (Fig. 6).

Additional factors related to predation may similarly cause encephalization and longevity to evolve independently. For example, carnivores exhibit large variation in body size (Fig. S1). Given that greater body size generally decreases predation risk (Cheney and Wrangham 1987; Isbell 1994), this creates varying constraints on extrinsic mortality across the clade. More specifically, while many large carnivores, such as bears, are apex predators (i.e., with no predators of their own), many small carnivores are subject to high levels of predation, especially from other carnivorans (Fedriani et al. 2000). Furthermore, many carnivoran species have developed noncognitive defenses against predators (e.g., musk production), which may relax selection on cognitive abilities (Fig. 5).

Finally, certain reproductive strategies may prevent the correlated evolution of encephalization and longevity. Although older comparative analyses limited to carnivorans found positive correlations between relative brain size and lifespan (Cutler 1979; Gittleman 1986), recent studies suggest that in carnivoran families with allomaternal care, relative brain size is actually negatively correlated with lifespan (Isler and van Schaik 2009a). In these species, helpers may reduce the cost of reproduction, allowing greater reproductive success relative to the expected reduction caused by larger relative brain size, thereby alleviating the need for a longer lifespan (Fig. 6; Isler and van Schaik 2009a). Furthermore, the cognitive benefits associated with larger brains may be translated solely into greater reproductive success, rather than increasing adult survival. Although relative brain size predicts problem-solving ability in carnivorans (Benson-Amram et al. 2016), the majority of the trials in this study (77%; Table S8) were run on individuals from families without allomaternal care,

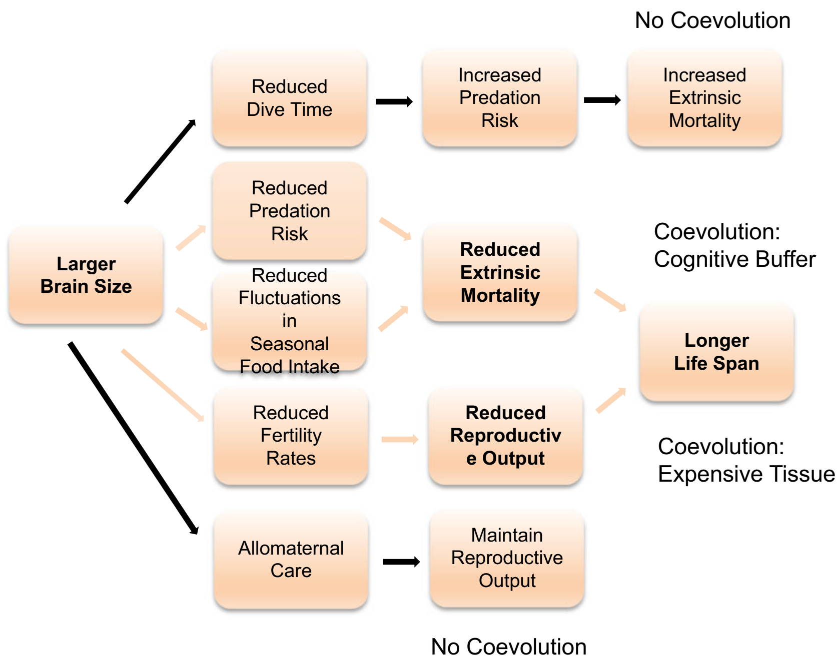

Figure 6. **Once relatively large brains evolve, various life history and ecological factors can either promote (in orange) or prevent (in black) the emergence of long lifespans.**

and a reexamination of the results indicates that families with helpers perform worse on problem-solving tasks than families without allomaternal care (Table S8).

While encephalization and longevity appear to evolve independently within multiple mammalian orders, primates, and rodents break from this pattern, albeit in different ways. This suggests that the delayed benefits and cognitive buffer/expensive brain hypotheses may be applicable to distinct instances of correlated evolution. It is important to note that these ideas are not mutually exclusive in the sense that it is not necessary that only one process governs the entire evolutionary history of these variables within a taxonomic group. Rather, they may act together to generate positive feedback favoring additional increases in both traits (Sol 2009a,b,). Accordingly, the results presented here reflect the process that likely governs the majority of the evolutionary history of these groups. Other transition pathways may also occur, but not often enough that their rates were significantly different from zero. It may also be the case that encephalization and longevity sometimes increased together too rapidly to detect any significant move through nonsignificant intermediate stages. Finally, it is possible that one trait increased at the origin of a group, thereby facilitating increases in the other trait throughout that group's subsequent evolutionary history.

Our finding that encephalization and longevity evolved in a correlated manner in primates is consistent with multiple studies reporting a positive correlation between relative brain size and maximum lifespan in this order (Austad and Fischer 1992; Hakeem et al. 1996; Deaner et al. 2003; Kaplan et al. 2003; Barrickman et al. 2008; Street et al. 2017). Given that encephalization appears to promote longevity within primates, the cognitive buffer (Deaner et al. 2003) and expensive brain hypotheses (Isler and van Schaik 2009a) may be most relevant to recent primate evolution, although the delayed benefits hypothesis might link observed patterns to the unique aspects of primate origins. In particular, our results suggest that relatively large brained, long-lived primate species evolved from species that already had relatively large brains. According to the cognitive buffer hypothesis, encephalized species are afforded cognitive abilities that make them better equipped to respond to ecological challenges, thereby reducing extrinsic mortality and increasing lifespan. This is in line with comparative studies of primate cognition, as relative brain size is positively correlated with foraging innovation (Reader and Laland 2002), technological innovation (Navarrete et al. 2016), nontechnical innovation (Navarrete et al. 2016), social learning (Reader and Laland 2002; Street et al. 2017), self-control (Maclean et al. 2014), and performance on various learning tasks (Passingham 1975; Deaner et al. 2007). Furthermore, primate species with relatively larger brains experience reduced seasonality in food intake, suggesting an ability to adjust foraging strategies during times of food scarcity (van Woerden et al. 2012).

Our results also suggest that primates do not tend to pass through a long-lived, small-brained intermediate stage when evolving from small-brained/short-lived to large-brained/long-lived (Fig. S2; Tables 3, S6–S7). In fact, only ~5% of primates are classified as long-lived and small-brained when using our standard cutoffs (relative to fellow primates; Table 1). These patterns are likely related to the unique aspects of primate locomotor and habitat evolution. Primates are long-lived among mammals generally, likely because they are mostly arboreal, which may improve their ability to flee predators, thereby reducing extrinsic mortality (Shattuck and Williams 2010). Arboreality appears to reduce extrinsic mortality in most mammalian orders, as arboreal species exhibit longer lifespans than terrestrial species of similar body sizes (Pomeroy 1990; Austad and Fischer 1992; Shattuck and Williams 2010; Healy et al. 2014). Within primates specifically, however, there is no significant difference in longevity between arboreal and terrestrial species (Shattuck and Williams 2010). Shattuck and Williams (2010) suggest this inconsistency may exist because arboreality is the primitive condition for primates (Dagosto 1988; Szalay 2007; Chester et al. 2017), while many other mammalian orders are characterized by terrestrial evolutionary histories, in which arboreality evolved multiple times (Haines 1958; Szalay 2007). Accordingly, if extended lifespans represent the ancestral primate condition, then most primates may live long enough to develop (Lefebvre et al. 2006) and benefit from (Deaner et al. 2003) the "delayed" benefits of complex cognitive abilities afforded by encephalization. In this case, the evolution of extended lifespans early in primate evolution may have promoted subsequent increases in encephalization throughout the order, thereby generating positive feedback and favoring additional increases in lifespan within some groups.

Rodents exhibit a different pattern from primates, in which highly encephalized, long-lived rodent species tend to evolve from species that already exhibit long lifespans. The majority of our models (varying by phylogeny or trait cutoff) suggest that large-brained, long-lived rodent species tend to evolve from species that already had long lives (Tables 3, S6–S7). We cannot definitively state whether or not relatively large brained, long-lived rodent species also tend to evolve from species that already exhibit relatively large brains due to the effects of phylogenetic uncertainty and categorization effects of different cutoff (Tables S6–S7). We nevertheless focus our discussion on the majority of our findings and the factors that may promote longevity preceding encephalization in this order. More specifically, this pattern may be related to certain aspects of rodent life histories and reproductive constraints. According to the delayed benefits hypothesis, investing in a large brain is less likely to pay off in animals with fast life histories than animals with slow life histories (Deaner et al. 2003). This may be particularly relevant for species at the very fast end of the life-history spectrum, such as rodents (Fig. S1), since investment in growth and development is already extremely limited. Consequently, even small increases in lifespan for short-lived rodent species might be enough to significantly increase the benefits of a relatively larger brain. Furthermore, increases in encephalization for very small-bodied mammals may only have modest impacts on mortality since smaller animals are subject to greater predation risk (Isbell 1994). Finally, the cognitive benefits afforded by encephalization may be translated into increased reproductive success rather than adult survival. Provisioning of offspring by both males and other group members evolved multiple times in rodents (Isler and van Schaik 2012), likely reducing the cost of reproduction of more encephalized offspring and lessening the need for a longer lifespan (Isler and van Schaik 2009a). Accordingly, provisioning rodent species exhibit relatively larger brains, but do not exhibit significantly longer lifespans (Isler and van Schaik 2012).

Given that the majority of species included in studies of encephalization and longevity (e.g., Gonzalez-Lagos et al. 2010) are primates and rodents, it is not surprising that such studies consistently conclude that these traits evolved in a correlated fashion across mammals. However, the results presented here highlight the dangers of combining very different groups of animals within broader comparative analyses, since the traits in question may exhibit markedly different evolutionary patterns within orders. To confirm that species sampling is not influencing overall results, supplementary analyses should be conducted within subgroups, especially in those for which previous research has questioned or rejected the evolutionary relationship under examination.

AUTHOR CONTRIBUTIONS

All authors designed the project and collected data. A.R.D. performed analyses and wrote the manuscript with input from N.A.T, S.A.W., and M.R.S.

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DATA ARCHIVING

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CONFLICT OF INTERESTS

The authors declare no competing interests.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1. Statistical relationship between: 1) brain and body size; and 2) lifespan and body size.
Table S2. Contingency tables and Fisher's exact test results showing the distribution of species across relative brain size and lifespan categories using our standard cutoffs.
Table S3. Contingency tables and Fisher's exact test results showing the distribution of species across relative brain size and lifespan categories using various cutoffs.
Table S4. Comparisons of independent and correlated models of evolution using our standard cutoffs.
Table S5. Comparisons of independent and correlated models of evolution using various cutoffs.
Table S6. Likelihood comparisons of unconstrained correlated evolution models (UCM) versus constrained models (CCM) in which one of the transitions between states has been forced to be zero using our standard cutoffs.
Table S7. Likelihood comparisons of unconstrained correlated evolution models (UCM) versus constrained models (CCM) in which one of the transitions between states has been forced to be zero using various cutoffs.
Table S8. Comparison of problem-solving ability (as measured by the percentage of successful trials) between families of mammalian carnivores with and without allomaternal care (data from Benson-Amram, et al., 2016).
Figure S1. Variation in lifespan (A), brain size (B), encephalization quotient (C), and body size (D) within and between mammalian groups.
Figure S2. Flow diagrams depicting correlated evolutionary patterns of relative brain size and lifespan in Primates, Rodentia, and Euarchontoglirans as presented in Table 3.